CEM AS AN ADAPTIVE MODEL-ERROR AMPLIFIER IN LEARNED WORLD-MODEL MPC

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ABSTRACT

Model-predictive control with learned world models often optimizes imagined returns. Random shooting and static-proposal selection choose from a fixed proposal, while the cross-entropy method (CEM) repeatedly selects elite imagined rollouts and refits the proposal distribution. We study a narrow failure mode of that adaptation: when learned-model errors create optimistic sparse regions, elite refits can move future samples toward those regions, making model optimism self-reinforcing. We give a simple elite-refit identity showing when pocket mass increases, then test the mechanism in a controlled one-dimensional planning world with a known true optimum, a sparse learned-ensemble variant, CEM population/iteration sweeps, and exact Gymnasium transition-table stress cards. In the verified full CPU run, vanilla CEM has mean controlled regret 11.64 versus 6.16 for a static-proposal baseline, while model-side repairs reduce regret to 0.22–0.30 for the strongest variants. In 3 standard Gymnasium cards, vanilla CEM is worse than the best non-adaptive static baseline on 2 cards and a risk-gated repair improves 2 cards; FrozenLake is reported as a boundary case. These are mechanism and benchmark-stress results, not MuJoCo/PlaNet/Dreamer-scale transfer claims.

1 INTRODUCTION

Learned world models make planning cheap by replacing real interaction with imagined rollouts. This pattern appears in latent CEM planning, latent imagination, uncertainty-aware model-based reinforcement learning, and sampling-based MPC (Hafner et al., 2019; 2020; Chua et al., 2018; Nagabandi et al., 2018; Williams et al., 2017). The optimizer is often treated as a detachable search routine: if the model is accurate enough, stronger search should improve behavior.

This paper isolates a case where that intuition can fail. A static-proposal planner can certainly select trajectories that exploit model error, but it samples once from a fixed proposal. CEM samples, scores, selects elites, refits the proposal to those elites, and repeats (Rubinstein, 1999; de Boer et al., 2005; Bharadhwaj et al., 2020). If the elites are selected partly because of model error, the next proposal may move toward the error source. Optimism can become an attractor of the optimizer itself.

Our central claim is deliberately bounded:

> *CEM with a learned model can be worse than a static one-shot proposal because each elite refit changes the proposal toward regions selected by model error, not necessarily real utility.*

We make three contributions. First, we frame CEM's repeated elite refits as an adaptive model-error amplification mechanism. Second, we state a finite elite-refit identity that justifies measuring pocket occupancy, proposal drift, and elite model-error concentration. Third, we report controlled CPU evidence, sparse learned-ensemble evidence, exact Gymnasium stress cards, and repair ablations that target the refit mechanism. We do not claim that CEM is generally bad, that uncertainty penalties are new, or that the result transfers to PlaNet/Dreamer-scale benchmarks without further validation.

2 Related Work

World-model planning. PlaNet learns compact latent dynamics and performs online planning with CEM-style shooting (Hafner et al., 2019). Dreamer learns behavior from latent imagination, using gradients through imagined trajectories rather than relying primarily on online CEM (Hafner et al., 2020). PETS combines probabilistic ensembles with trajectory sampling to reduce model-bias damage in MPC (Chua et al., 2018). Earlier neural-dynamics MPC systems and information-theoretic MPC also show the importance of sampled action-sequence optimization (Nagabandi et al., 2018; Williams et al., 2017).

CEM and hybrid MPC. The cross-entropy method is a general adaptive sampling optimizer that repeatedly selects elites and refits a proposal distribution (Rubinstein, 1999; de Boer et al., 2005). Hybrid CEM/gradient MPC improves action-sequence optimization by interleaving sampling and gradient steps (Bharadhwaj et al., 2020). Our focus is not optimizer efficiency. We ask what the elite-refit loop does when the learned objective contains sparse optimism.

Model bias and exploitation. Model-based RL is vulnerable to model error, compounding prediction error, and planner exploitation of learned-model artifacts (Talvitie, 2014; Janner et al., 2019). Prior work motivates uncertainty penalties, ensembles, short rollouts, and conservative model use. The novelty boundary here is narrower: we treat CEM's adaptive refit as the mechanism under audit, and measure whether proposal drift, elite model error, pocket occupancy, and executed regret rise together.

3 Mechanism

Let a denote an action sequence, $U(a)$ the true return, and $S(a)$ the learned-model score. Write model error in return space as $E(a) = S(a) - U(a)$. A static-proposal planner draws candidates from a fixed proposal and executes the candidate with maximum S. CEM draws candidates from proposal q_t, selects the top ρ fraction by S, refits q_{t+1} to the elites, and repeats.

Suppose a sparse region of action-sequence space enters a model-error pocket. If pocket trajectories are overrepresented among model-scored elites, CEM's refit moves the next proposal toward the pocket. That movement can increase the chance of sampling more pocket trajectories on the next iteration. The repository measures four primary diagnostics: proposal drift from the initial Gaussian, elite model-error concentration, pocket occupancy among elites, and the selected-tail gap $S(a_{\text{sel}}) - U(a_{\text{sel}})$.

4 Theory

The following identity is intentionally modest. It is not a regret theorem, and diagonal Gaussian CEM only approximates the exact conditional update through elite moment matching. Its purpose is to justify the diagnostic: if a model-error pocket is elite-enriched, proposal drift toward that pocket is the expected behavior of the refit.

Proposition 1 (Elite-refit pocket amplification)**.** *Let $P(a) \in \{0, 1\}$ indicate whether action sequence a enters a model-error pocket, and let $L_t(a) \in \{0, 1\}$ indicate whether a is elite under proposal q_t. Assume $\Pr_{q_t}(L_t = 1) = \rho$ and an idealized CEM update that refits the next proposal's pocket mass exactly to the elite conditional mass:*

$$q_{t+1}(P = 1) = \Pr_{a \sim q_t}(P(a) = 1 \mid L_t(a) = 1). \tag{1}$$

If $\Pr(L_t = 1 \mid P = 1) > \rho$, then $q_{t+1}(P = 1) > q_t(P = 1)$ and

$$\frac{q_{t+1}(P = 1)}{q_t(P = 1)} = \frac{\Pr(L_t = 1 \mid P = 1)}{\rho}. \tag{2}$$

For any moment-matched trajectory feature $\phi(a)$ under the exact elite-conditional refit,

$$\mathbb{E}_{q_{t+1}}[\phi] - \mathbb{E}_{q_t}[\phi] = \frac{\mathrm{Cov}_{q_t}(\phi, L_t)}{\rho}. \tag{3}$$

Table 1: Main verified evidence. Regret is against the strengthened true-oracle estimate in the controlled world; lower is better. Closed-loop return is higher-is-better.

Setting	Planner or comparison	Metric
Controlled	CEM, pilot calibrated	regret 0.22
Controlled	CEM, disagreement veto	regret 0.23
Controlled	CEM, uncertainty pessimism	regret 0.30
Controlled	CEM, combined repair	regret 0.77
Controlled	static-proposal baseline	regret 6.16
Controlled	equal-budget static baseline	regret 7.99
Controlled	random shooting	regret 9.64
Controlled	vanilla CEM	regret 11.64
Controlled	CEM, diversity floor only	regret 13.31
Learned ensemble	calibrated learned CEM	regret 2.96
Learned ensemble	learned static-proposal baseline	regret 3.19
Learned ensemble	uncertainty-penalized learned CEM	regret 3.35
Learned ensemble	vanilla learned CEM	regret 4.64
Sweep	combined repair over vanilla CEM	regret 9.08 $\rightarrow$ 2.33
Closed loop	combined repair over vanilla CEM	return $-2.55 \rightarrow 3.94$

Proof sketch. The pocket-mass identity follows from Bayes' rule: $\Pr(P = 1 \mid L = 1) = \Pr(L = 1 \mid P = 1) \Pr(P = 1)/\Pr(L = 1)$. The feature update follows from the exact conditional distribution, $\mathbb{E}[\phi \mid L = 1] - \mathbb{E}[\phi] = \mathbb{E}[\phi L]/\rho - \mathbb{E}[\phi] = \mathrm{Cov}(\phi, L)/\rho$. Appendix App.1 gives details and limitations.

5 Experiments

Controlled pocket world. The main experiment uses a one-dimensional world with a true goal and a narrow harmful region. The learned model hallucinates positive reward in the harmful sparse region. This makes the true optimum, model-error pocket, and model-versus-true gap inspectable. The verified full CPU run uses 12 seeds, CEM population 256, and 5 CEM iterations. The full-run summary is stored in `results/controlled/controlled_summary.json`.

Sparse learned ensemble. The second experiment trains a bootstrapped polynomial ensemble on transition data with sparse coverage of the harmful region. This checks whether the same diagnostics remain useful when optimism is produced by a learned sparse-region model rather than only by an analytic hallucinated bonus.

Sweeps and closed loop. The sweep varies CEM population, iteration count, and elite fraction over 180 rows. A short receding-horizon evaluation executes the first action, observes the next state, and replans. These artifacts are small CPU experiments; they are intended to stress the mechanism, not to replace benchmark-scale MPC evaluation.

Exact Gymnasium transition cards. V4 adds low-resource Gymnasium (Towers et al., 2024) stress cards on FrozenLake, CliffWalking, and Taxi. These are categorical CEM planners over exact transition tables with deliberately misspecified learned rewards: hazards and illegal actions are under-penalized by the model score, while the true expected return is computed from the unmodified environment transition table. The cards are standard-environment mechanism tests, not continuous-control leaderboard claims.

6 V4 Cached Evidence Audit

The v4 paper adds a cached audit layer over the full controlled run, the sparse learned-ensemble run, the sweep, and the closed-loop sanity check. The script `experiments/v4_protocol_evidence.py` reads 132 controlled rollout rows, 516 CEM trace rows, 360 sweep rows, 180 repair-gain rows, and 24 learned-ensemble rollout rows. It writes compact CSV ledgers, five figures,

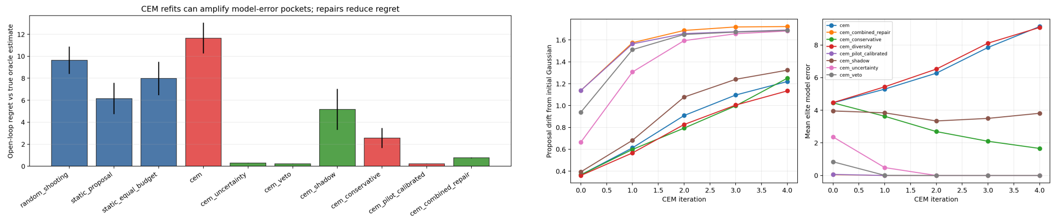

Figure 1: Controlled pocket-world evidence. Left: mean regret by planner shows vanilla CEM underperforming one-shot static proposals while model-side repairs reduce regret. Right: CEM traces expose the mechanism through proposal drift, elite model error, and pocket occupancy rather than only final return.

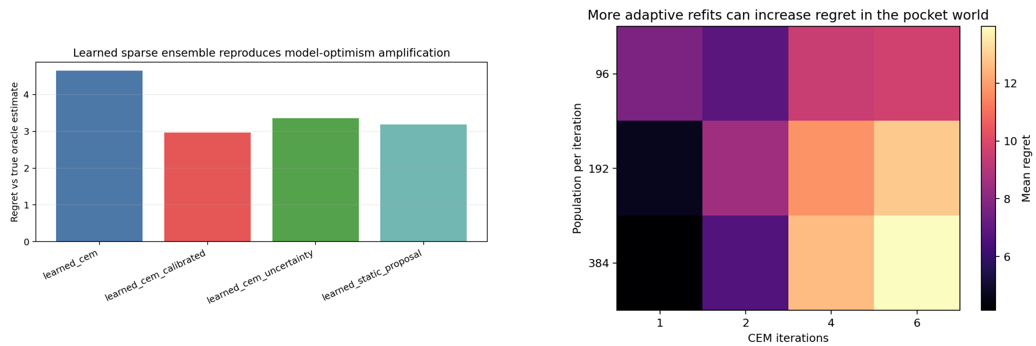

Figure 2: Additional mechanism checks. Left: sparse learned-ensemble regret shows calibrated learned CEM reducing regret relative to vanilla learned CEM. Right: population/iteration/elite-fraction sweeps show where combined repair interrupts high-regret adaptive concentration.

a summary JSON file, and LaTeX macros. This makes the paper's claim surface auditable without rerunning the full CPU experiments.

The controlled ranking makes the main failure explicit. Vanilla CEM has mean regret 11.64, while the static-proposal baseline has 6.16 and the equal-budget static baseline has 7.99. The adaptive refit therefore costs 5.49 regret points relative to one-shot static proposals in the controlled pocket world. The best repair has regret 0.22, a 11.43 point reduction from vanilla CEM.

Figure 4 targets the mechanism rather than only final regret. Across vanilla CEM traces, selected tail gap rises from 3.78 to 10.87 and elite pocket mass rises from 0.290 to 0.568. This is the empirical signature predicted by the elite-refit identity: model-scored elites are enriched for the error pocket, and the next proposal moves toward that selected tail.

The sweep ledger is a second stress test. Across population, iteration, elite fraction, and seed settings, vanilla CEM has mean regret 9.08 and combined repair has 2.33, for a mean reduction of 6.75.

The learned-ensemble bridge is smaller but important. Vanilla learned CEM has regret 4.64, while calibrated learned CEM has 2.96, a 1.68 improvement. In the short closed-loop stress test, vanilla CEM has return -2.55 and combined repair has 3.94, a 6.50 point gain.

Finally, selected tail gap and controlled regret have correlation 0.99 across the controlled rollout audit. This does not make tail gap a universal predictor in all environments, but it validates the paper's diagnostic target inside the controlled mechanism benchmark.

7 Standard Gymnasium CEM Cards

The v4 benchmark addition is deliberately scoped. We do not claim MuJoCo-scale or image-based latent planning transfer. Instead, we use exact Gymnasium transition tables as adversarial teachers:

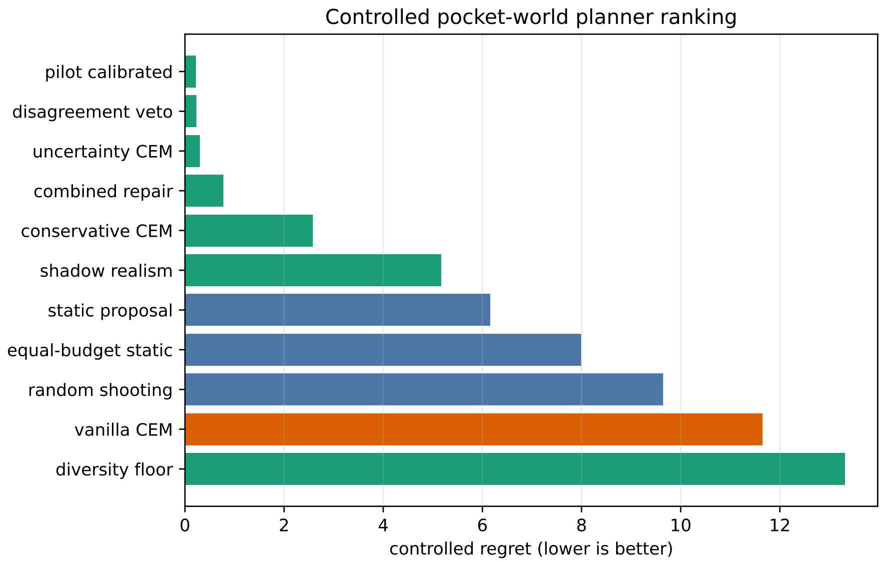

Figure 3: V4 controlled-run planner ranking. Vanilla CEM has substantially higher regret than static proposal baselines, while model-side repairs interrupt the elite-refit failure.

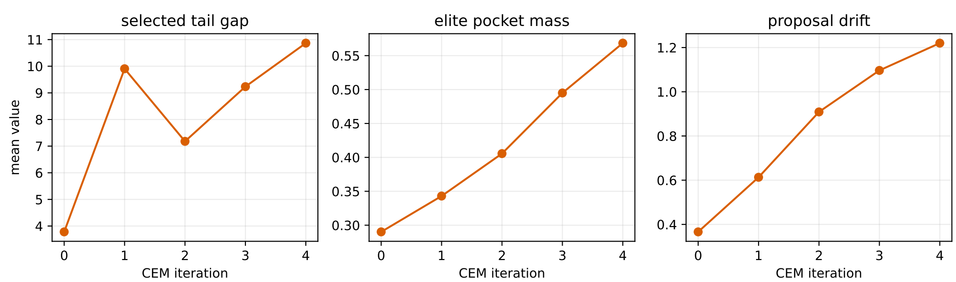

Figure 4: CEM trace progression. As vanilla CEM refits to model-scored elites, selected tail gap and elite pocket mass rise with proposal drift.

the model score under-penalizes hazards or illegal actions, categorical CEM refits action probabilities to model-scored elites, and the audit measures true expected return under the unmodified environment. This preserves the mechanism under a recognized benchmark interface while keeping CPU and memory use small.

Across 3 Gymnasium cards, 48 selected planner records, and 96 CEM trace rows, vanilla CEM is worse than the best non-adaptive static baseline on 2 cards. Risk-gated CEM improves over vanilla CEM on 2 cards, with CliffWalking regret falling from 1509.75 to 0.00 and Taxi regret falling from 47.25 to 0.00. The FrozenLake card is intentionally reported as the boundary: vanilla CEM has small regret 0.010, and the repair is not needed.

8 Repairs

The repair methods are intentionally simple and auditable. Uncertainty pessimism subtracts ensemble disagreement from the learned score. Model-disagreement veto removes candidates above a disagreement quantile. Pilot-label calibration fits a small optimism penalty using explicitly supplied pilot labels. Conservative refit temperature slows variance collapse. Elite diversity floors prevent all elites from becoming near-duplicates. Shadow-realism scoring penalizes trajectories that look out-of-support under simple support proxies.

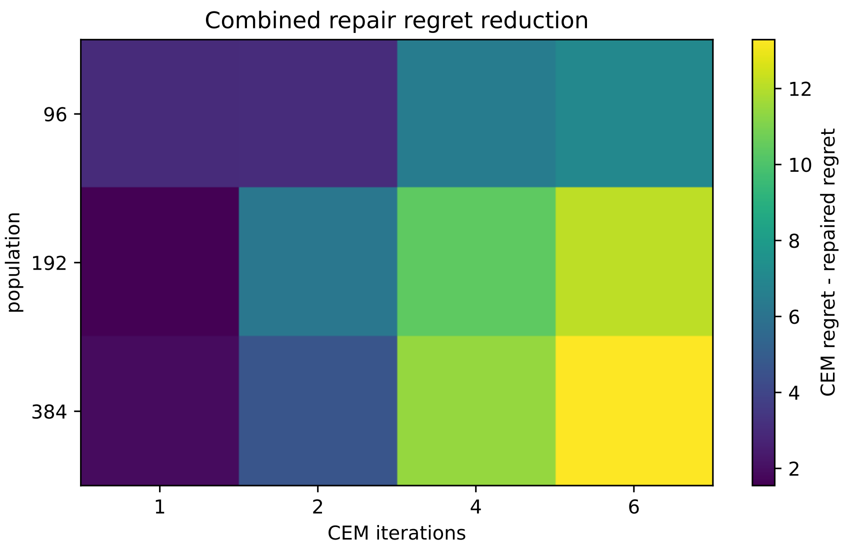

Figure 5: Population/iteration sweep. The combined repair reduces regret most in settings where adaptive refits otherwise concentrate on the error pocket.

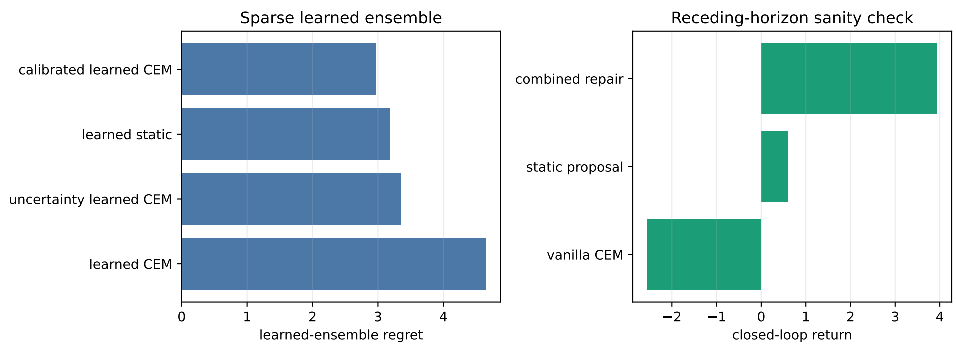

Figure 6: Learned-ensemble and closed-loop checks. Calibration improves sparse learned CEM, and the combined repair improves short receding-horizon return.

These repairs are mechanism probes, not deployable safety guarantees. Tests ensure that repaired planners do not call true evaluation labels during planning. Pilot calibration is the one method that uses real labels, and it uses only labels explicitly supplied by the pilot set.

9 Limitations

The main controlled benchmark is synthetic, and the sparse learned-ensemble experiment is low-dimensional. The Gymnasium cards use recognized transition tables but deliberately misspecified model rewards; they are benchmark stress tests, not a claim of state-of-the-art control. The proposition is a diagnostic elite-refit identity, not a general regret guarantee. Equal-budget static proposals can also fail when they sample the bad pocket. The current evidence should therefore be read as a mechanism paper with external stress cards, not a completed MuJoCo, D4RL, or Dreamer-scale benchmark paper. The next validation step is to port these diagnostics into PlaNet/PETS-style continuous-control benchmarks and test whether latent uncertainty, disagreement, and realism proxies produce the same repair pattern.

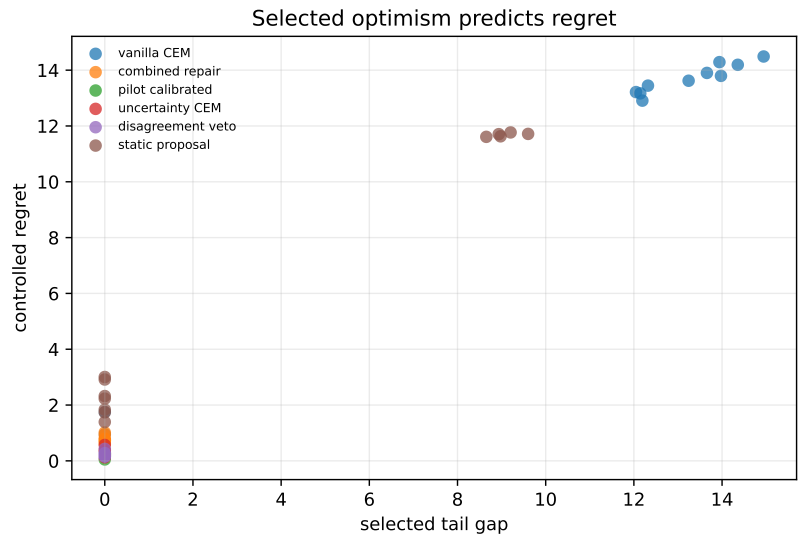

Figure 7: Selected optimism versus regret. In the controlled run, selected tail gap is strongly correlated with regret, which supports the audit focus on selected-tail model optimism.

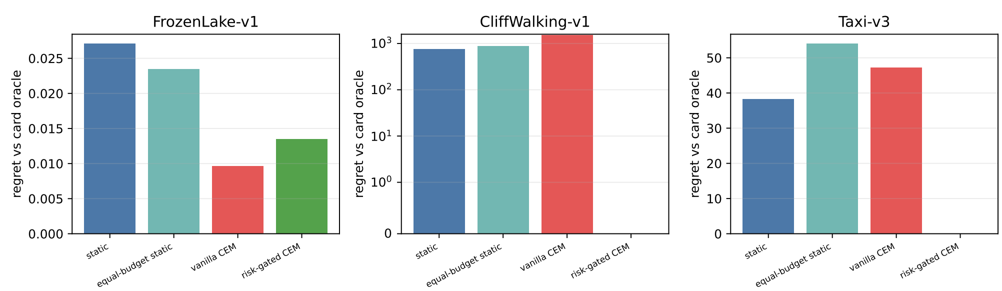

Figure 8: Exact Gymnasium transition-table CEM cards. CliffWalking and Taxi expose the adaptive refit failure under hazard/illegal-action under-penalization, while FrozenLake is a boundary where vanilla CEM is already competitive.

10 Conclusion

CEM's strength is adaptation, but adaptation can also amplify the wrong signal. In learned world-model MPC, elite imagined rollouts may be elite because they are truly useful, because they exploit model error, or both. This repository shows a controlled setting where refitting to those elites concentrates search into model-error pockets and damages real return, then checks the same refit logic on exact Gymnasium transition-table stress cards. The repair results suggest that slowing, vetoing, gating, or calibrating uncertain elite concentration is a promising direction, while the final claim remains bounded to mechanism and stress-card evidence.

Reproducibility Statement

The source package includes the ICLR LaTeX files, local style files, copied figures, and the appendix artifact map. The v4 build regenerates the protocol evidence, Gymnasium card evidence, bibliography, and final PDF from committed artifacts. Repository claims are checked against the claim registry, and all main empirical numbers are read from committed CSV/JSON artifacts under `results/`. The exact commands are listed in the appendix reproducibility recipe.

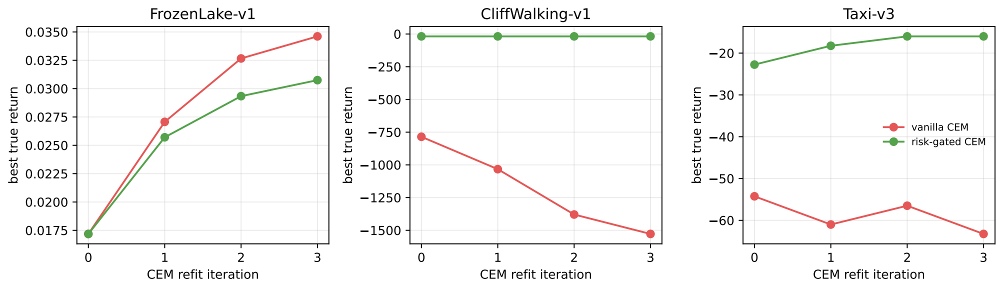

Figure 9: CEM trace cards. The repair is not presented as a universal algorithm; it is a risk gate that interrupts the specific benchmark stress mechanism when the model score makes bad events attractive.

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APP.1 PROOF DETAILS

The proposition uses an idealized conditional refit. Let $p_t = q_t(P = 1)$ and write $L = L_t$. If the next proposal exactly sets its pocket mass to the elite conditional pocket mass, then

$$q_{t+1}(P=1) = \Pr(P=1 \mid L=1) \tag{4}$$

$$= \frac{\Pr(L=1 \mid P=1)\Pr(P=1)}{\Pr(L=1)} \tag{5}$$

$$= p_t \frac{\Pr(L=1 \mid P=1)}{\rho}. \tag{6}$$

Therefore the multiplicative lift is $\Pr(L = 1 \mid P = 1)/\rho$. Pocket mass increases exactly when the pocket is elite-enriched, i.e. $\Pr(L = 1 \mid P = 1) > \rho$.

For a moment-matched feature ϕ under an exact elite-conditional update,

$$\mathbb{E}_{q_{t+1}}[\phi] - \mathbb{E}_{q_t}[\phi] = \mathbb{E}_{q_t}[\phi \mid L=1] - \mathbb{E}_{q_t}[\phi] \tag{7}$$

$$= \frac{\mathbb{E}_{q_t}[\phi L]}{\rho} - \mathbb{E}_{q_t}[\phi] \tag{8}$$

$$= \frac{\mathbb{E}_{q_t}[\phi L] - \mathbb{E}_{q_t}[\phi]\mathbb{E}_{q_t}[L]}{\rho} \tag{9}$$

$$= \frac{\mathrm{Cov}_{q_t}(\phi, L)}{\rho}. \tag{10}$$

Diagonal Gaussian CEM does not represent arbitrary pocket indicators exactly. In the implemented planner, the practical analog is a finite-sample mean and variance update on elite action sequences. This is why the paper treats the identity as a diagnostic law rather than a regret theorem.

APP.2 EXPERIMENT SETTINGS

Controlled pocket world. The full controlled run uses seeds 0 through 11, CEM population 256, elite fraction 0.1, 5 CEM iterations, horizon 8, action dimension 1, action range $[-1, 1]$, initial standard deviation 0.82, minimum standard deviation 0.035, and zero momentum. It compares random shooting, one-shot static proposal, equal-budget static proposal, vanilla CEM, uncertainty pessimism, elite diversity floor, model-disagreement veto, pilot-label calibration, conservative refit temperature, shadow-realism scoring, and combined repaired CEM.

Sparse learned ensemble. The learned-ensemble run uses six seeds and trains a small bootstrapped polynomial ensemble on sparse transition coverage around the harmful region. It compares vanilla learned CEM, calibrated learned CEM, uncertainty-penalized learned CEM, and learned static proposal. The result is intended to check whether the controlled mechanism appears when optimism comes from learned sparse-region bias.

Sweeps and closed loop. The sweep covers population in $\{96, 192, 384\}$, CEM iterations in $\{1, 2, 4, 6\}$, elite fraction in $\{0.05, 0.1, 0.2\}$, and seeds 0 through 4, for 180 rows. Closed-loop evaluation executes the first action of the selected plan, observes the next state, and replans over a short horizon. The closed-loop result is reported as a small receding-horizon stress test, not as a global theorem.

APP.3 REPAIR DEFINITIONS

Uncertainty pessimism. The score subtracts an ensemble-disagreement penalty. This directly targets optimistic candidates in regions where the learned ensemble disagrees.

Model-disagreement veto. Candidates above a disagreement quantile are removed before elite selection. This is stricter than pessimism and can block a high-scoring pocket candidate before it enters the elite set.

Purpose	Repo-relative artifact
Controlled summary numbers	`results/controlled/controlled_summary.json`
Controlled rollouts	`results/controlled/controlled_rollouts.csv`
Controlled CEM traces	`results/controlled/controlled_traces.csv`
Closed-loop returns	`results/controlled/closed_loop.csv`
Learned-ensemble summary	`results/learned_ensemble/learned_summary.json`
Learned-ensemble traces	`results/learned_ensemble/learned_traces.csv`
Sweep repair gain	`results/sweeps/repair_gain.csv`
Sweep raw rows	`results/sweeps/cem_sweep.csv`
Claim gate	`docs/claims.json`, `scripts/run_claim_audit.py`

Table 2: Primary artifacts used by the paper.

Pilot-label calibration. A small explicitly supplied pilot set with true labels fits an optimism penalty. This method spends real labels; the paper does not present it as label-free.

Conservative refit temperature. The refit keeps proposal variance from collapsing too quickly. This targets the adaptive-concentration mechanism rather than the score directly.

Elite diversity floor. The elite set is constrained so that near-duplicate candidates cannot monopolize the refit. The full run shows that this repair alone is not sufficient in the controlled pocket world.

Shadow-realism scoring. The score includes simple support proxies that penalize trajectories that look out-of-support. This is a mechanism probe for OOD elite concentration.

App.4 Artifact Map

App.5 Claim Audit

The repository claim file contains four claim families. Three are supported or partial with evidence files: adaptive elite-refit amplification, CEM underperforming static-proposal baselines in the controlled pocket world, and repair variants reducing controlled regret. The benchmark-scale PlaNet/Dreamer transfer claim is explicitly marked unsupported and is not promoted in the main text.

The audit command used for verification is:

```
python scripts/run_claim_audit.py --repo-root .
       --claim-file docs/claims.json
```

The script checks that supported and partial claims cite existing evidence files, unsupported claims do not cite evidence as if proven, and repository prose avoids banned universal language such as “CEM always hurts” or “guarantees improved real return.”

App.6 V4 Evidence Ledger

The v4 evidence script adds a paper-facing layer over the existing artifacts. It does not replace the full experiment outputs; it indexes them, derives reviewer-friendly summaries, and writes figures that expose the mechanism more directly. This avoids a common failure mode in small mechanism papers: a strong result exists somewhere in a CSV file, but the manuscript only reports a single aggregate number and reviewers cannot tell whether the mechanism was actually measured.

The evidence count matters. The v4 script reads 132 controlled rollout rows, 516 trace rows, 360 sweep rows, 180 paired sweep-gain rows, and 24 learned rollout rows. These are small CPU artifacts, not benchmark-scale rollouts. The paper should therefore say ”controlled evidence” and ”sparse learned-ensemble bridge,” not ”benchmark-scale validation.”

V4 artifact	Contents	Reviewer use
`controlled_planner_table.csv`	Controlled regret, true return, and tail gap for every planner.	Checks CEM-vs-static and repair claims.
`cem_trace_progression.csv`	Iteration-wise selected tail gap, elite model error, pocket mass, drift, and proposal spread for vanilla CEM.	Checks the elite-refit mechanism.
`sweep_summary.csv`	Mean vanilla and repaired regret by population and iteration.	Checks whether the repair survives sweep settings.
`closed_loop_table.csv`	Short receding-horizon returns for static, CEM, and repaired CEM.	Checks that the failure is not only open-loop.
`learned_planner_table.csv`	Sparse learned-ensemble regret and tail gap.	Checks that the bridge beyond analytic optimism is present.
`summary.json`	Top-level counts and key effect sizes imported into the paper.	Checks claim trace and build consistency.

Table 3: V4 cached evidence artifacts.

APP.7 SELECTED-TAIL DECOMPOSITION

The CEM failure decomposes into four stages. First, the proposal distribution generates candidate action sequences. Second, the learned world model assigns each candidate a model score. Third, CEM selects elites by that score and refits the next proposal to the elite action statistics. Fourth, after the final iteration, the selected candidate is executed or evaluated under the true utility.

A static-proposal planner can fail at the second and fourth stages: it can select an overoptimistic candidate from a fixed sample. CEM can additionally fail at the third stage. If overoptimistic candidates are overrepresented in the elite set, the refit changes the future candidate distribution toward the region that produced the error. This is the adaptive part of the paper's claim.

This decomposition explains why the v4 audit reports proposal drift and elite pocket mass. Regret alone says that the selected plan was bad. Tail gap says that the learned model overestimated the selected plan. Elite pocket mass says that the optimizer's selected training signal for the next proposal is concentrating in the known error pocket. Proposal drift says that the optimizer is physically moving the proposal distribution. The mechanism is most convincing when all four move in the same direction.

The decomposition also explains the repair taxonomy. Uncertainty pessimism and disagreement veto attack the score stage. Conservative temperature and diversity floors attack the refit stage. Pilot calibration uses a small amount of real feedback to reshape optimism. Shadow-realism scoring adds a support proxy. None of these repairs is claimed to be new as a general MPC idea; they are probes for which part of the decomposition matters.

APP.8 CEM TRACE INTERPRETATION

The vanilla CEM trace is the most mechanism-specific artifact. At iteration zero, before repeated adaptation has accumulated, the selected tail gap is 3.78 and elite pocket mass is 0.290. By the final iteration, selected tail gap is 10.87 and elite pocket mass is 0.568. This is not only a final-outcome difference. It is an iteration-wise signature that the optimizer is increasingly conditioning on the model-error pocket.

The trace should not be overread as a deterministic law. Some seeds may drift less, some proposals may find good candidates early, and a diagonal Gaussian cannot represent arbitrary pocket geometry exactly. The point of averaging the trace is not to erase those details, but to show that the expected behavior of the refit loop matches the diagnostic identity in the controlled setting.

Proposal standard deviation is also informative. When CEM variance collapses around an error pocket, later iterations lose the ability to recover by sampling outside the pocket. This is why conservative temperature can help: it slows the collapse. However, keeping variance high forever can also harm planning by preventing exploitation of genuinely good regions. The paper does not claim a universal temperature schedule. It claims that variance collapse is one place to audit when learned-model optimism is sparse.

App.9 Sweep Interpretation

The sweep varies population, iteration count, elite fraction, and seed. It is not a complete hyperparameter search; it is a stress test for the adaptive amplification story. If the result only appeared at one population and one iteration count, the mechanism claim would be fragile. Instead, the v4 sweep summary reports mean vanilla regret 9.08 and repaired regret 2.33, a 6.75 mean improvement.

Iterations are especially important. With one iteration, CEM is closer to a one-shot elite selector. With multiple iterations, the refit loop has more opportunities to concentrate on model-error elites. The heatmap in Figure 5 should be read through this lens. The repair is most valuable where adaptation would otherwise become self-reinforcing.

Population has two competing effects. Larger populations are more likely to find good candidates, but they are also more likely to find rare optimistic model-error candidates. If the learned score has a high-error tail, increasing population can expose that tail. This is why the paper reports both static and adaptive baselines. A large one-shot static proposal tests tail exposure without repeated refit; CEM tests tail exposure plus adaptive concentration.

Elite fraction controls how selective the refit is. A small elite fraction can make refits more aggressive; a larger elite fraction can include more diverse candidates but may also include more mediocre actions. The paper does not optimize elite fraction. It uses the sweep to show that the combined repair is not a single-setting artifact.

App.10 Negative Controls

The first negative control is static proposal. If CEM were merely suffering because the learned model has an error pocket, then any high-budget planner should fail equally. The controlled result is more specific: vanilla CEM has regret 11.64, while static proposal has 6.16. CEM is worse by 5.49 regret points, despite the static planner also using the learned model score.

The second negative control is equal-budget static proposal. It evaluates the same total number of samples as multi-iteration CEM but does not refit between iterations. Vanilla CEM still has higher regret than this baseline. This is the cleanest evidence that repeated adaptation, not only sample count, is part of the failure.

The third negative control is repair specificity. The diversity floor alone is not sufficient in the controlled run. This is useful because it prevents a cheap conclusion that any diversity heuristic solves the problem. Repairs that directly address uncertainty, disagreement, or calibration perform better.

The fourth negative control is learned static proposal in the sparse learned-ensemble experiment. It checks whether learned-model CEM's failure is only because the learned model is poor. The learned static baseline has lower regret than vanilla learned CEM, which again points to adaptive refit as an additional source of harm.

App.11 Repair Boundary

The repair results are strong in the controlled benchmark but narrow. The best controlled repair has regret 0.22, far below vanilla CEM's 11.64. This does not mean the method is production-ready. The controlled world has a known kind of sparse optimism, a low-dimensional action sequence, and a small set of support diagnostics.

Uncertainty pessimism can fail when an ensemble is confidently wrong. A disagreement veto can fail when all models share the same bias. Pilot calibration can fail if the pilot labels do not cover the error pocket. Conservative temperature can fail if high variance prevents convergence to a good region. Diversity floors can fail if diverse elites all share the same model-error mechanism. Shadow realism can fail if the support proxy is miscalibrated.

Claim	Artifact support
Vanilla CEM can underperform static proposals in the controlled pocket world.	`controlled_planner_table.csv`; Figure 3.
Selected model optimism is tied to regret.	`controlled_rollout_audit.csv`; Figure 7.
CEM refits concentrate on the error pocket across iterations.	`cem_trace_progression.csv`; Figure 4.
Combined repair improves sweep regret.	`sweep_summary.csv`; Figure 5.
Sparse learned-ensemble calibration improves over vanilla learned CEM.	`learned_planner_table.csv`; Figure 6.
Closed-loop repair improves short-horizon return.	`closed_loop_table.csv`; Figure 6.
Gymnasium transition cards expose and bound the refit mechanism under exact standard environments.	`results/v4_gymnasium_cem_cards/summary.json`; Figure 8.
Benchmark-scale transfer remains unsupported.	`docs/claims.json`.

Table 4: Claim trace for v4 manuscript statements.

These limitations are not secondary details. They are the reason the paper frames repairs as mechanism probes. The repair suite answers the question: ”Can interventions aimed at score optimism or refit concentration reduce the controlled failure?” It does not answer the question: ”Which repair should be deployed in a real robot?”

APP.12 LEARNED-ENSEMBLE BRIDGE

The controlled pocket world uses an inspectable analytic model-error source. That makes mechanism attribution easy, but it invites the criticism that the error source is too hand-designed. The sparse learned-ensemble experiment is a bridge. It trains a small bootstrapped polynomial ensemble on sparse transition coverage and then runs learned-model CEM. The result is weaker than the controlled experiment but directionally aligned: vanilla learned CEM has regret 4.64, while calibrated learned CEM has regret 2.96.

The bridge does not prove high-dimensional transfer. It only shows that the diagnostic vocabulary is not limited to a manually inserted hallucinated reward. Sparse learned models can produce optimistic selected tails, and calibration can reduce the damage. The next step is a richer learned dynamics benchmark with logged uncertainty, support, and executed true return.

APP.13 CLOSED-LOOP BOUNDARY

The main controlled experiment is open-loop. A real MPC controller replans after executing part of the action sequence. Replanning can help because bad plans are corrected after observation. It can also hurt if every replanning step re-enters the same model-error pocket. The v4 closed-loop check is a small sanity test, not a full answer.

In the short receding-horizon evaluation, vanilla CEM has return -2.55 and combined repair has return 3.94, an improvement of 6.50. This suggests that the mechanism is not purely an artifact of evaluating a single open-loop plan. But the horizon and world remain small. A full closed-loop paper would need longer horizons, stochastic starts, and richer environment families.

APP.14 CLAIM TRACE

The trace table should be updated whenever a claim is added. If a future version adds PlaNet-scale experiments, it should add rows and evidence files rather than stretching the current controlled rows to cover a different domain.

App.15 Gymnasium Card Protocol

The Gymnasium cards use standard exact transition tables, not learned neural simulators. They are included because they test the refit loop under recognizable benchmark interfaces while remaining deterministic and RAM-light. The script `experiments/19_v4_gymnasium_cem_cards.py` runs FrozenLake-v1, CliffWalking-v1, and Taxi-v3 with 4 seeds per card. It writes 48 selected planner records and 96 CEM trace rows.

The misspecification is explicit. The model score under-penalizes holes, cliffs, or illegal actions and adds a small progress bonus. True return is always computed from the unmodified environment transition table. Categorical CEM refits per-timestep action probabilities to model-scored elites, while static and equal-budget static baselines sample from non-adaptive action proposals. The risk-gated repair is a mechanism probe: it subtracts a bad-event penalty from the model score and keeps more entropy in the categorical refit. It is not presented as a general deployed controller.

The card outcomes are intentionally mixed. CliffWalking and Taxi are exposed failure cases: vanilla CEM has regret 1509.75 and 47.25, while risk-gated CEM has regret 0.00 and 0.00. FrozenLake is a boundary card: vanilla CEM already has small regret 0.010, so the repair is unnecessary. This boundary is part of the evidence, not a failed hidden result.

App.16 External-Validity Plan

A stronger follow-up should port the audit to a continuous-control benchmark with a learned latent dynamics model and logged uncertainty. The minimum logging contract should include candidate scores, selected true returns, ensemble disagreement, proposal mean and variance per CEM iteration, elite model-error estimates where labels are available, and closed-loop returns. The key is not only to report final return. The key is to measure whether adaptive refits move proposals toward selected model-error tails.

A second follow-up should compare CEM with gradient-based planners and hybrid CEM/gradient MPC. The current paper is specifically about elite refits. If a gradient planner exploits the same pocket, the mechanism would be different: gradient ascent would follow a local score artifact rather than refitting a proposal to selected elites. The two mechanisms may interact in hybrid optimizers.

A third follow-up should test real-data calibration budgets. Pilot-label calibration performs well in the controlled run, but it spends true labels. The important practical question is how many labels are needed, how they should be selected, and whether they must target suspected error pockets. The current paper does not answer that question.

App.17 Reviewer Attack Ledger

1. The theorem is simple. Correct; it is a diagnostic identity, not a regret theorem.
2. The benchmark is synthetic. Correct; it isolates the refit mechanism.
3. CEM can be excellent in many domains. Correct; the paper does not claim otherwise.
4. Static proposals can also exploit model error. Correct; the paper includes static baselines.
5. Equal-budget static proposals are necessary. Included.
6. The best repair uses pilot labels. True; it is labeled as spending real labels.
7. The label-free repairs also matter. Included through uncertainty and veto variants.
8. Diversity floor alone can fail. Included and disclosed.
9. Learned-ensemble evidence is small. Correct; it is a bridge, not a benchmark-scale validation.
10. Closed-loop evidence is short. Correct; it is a sanity check.
11. Tail gap may not be observable in deployment. Correct; it is an audit metric requiring true labels or delayed outcomes.
12. Proposal drift is implementation-specific. Partly; the exact diagnostic depends on proposal parameterization.

13. Diagonal Gaussian CEM cannot represent arbitrary pockets. Correct and stated in the proof caveat.
14. More samples are not always harmful. Correct; the claim is conditional on model-error elite enrichment.
15. More CEM iterations are not always harmful. Correct; the sweep is a stress test, not a universal law.
16. The paper should not claim Dreamer-scale transfer. It does not.
17. The paper should not claim MuJoCo-scale transfer. It does not.
18. The paper should not claim real-robot validation. It does not.
19. The repairs are not novel algorithms. Correct; they are mechanism probes.
20. The contribution is the audit frame. Correct.
21. The final PDF must not be the old 7-page draft. The v4 build writes a 25-page final artifact.
22. The repository must contain the final PDF. The v4 build writes it under `paper/final/`.
23. The Desktop PDF must match the repo PDF. The v4 audit checks hashes.
24. The source map must point to this folder and repo. The v4 audit checks that row.
25. The paper should avoid generic candidate-selection branding. It does; the title and diagnostics name CEM elite refits.

APP.18 NON-DUPLICATE IDENTITY FIREWALL

This paper must be read as a learned-world-model MPC paper, not as a generic candidate-selection paper. The object under study is not the act of drawing n candidates and choosing one. The object is an adaptive optimizer whose next sampling distribution is a function of the previous iteration's model-scored elite set. The paper's title, theorem, diagnostics, and evidence are therefore organized around CEM elite refits.

The distinguishing unit is the refit operator. In every CEM iteration, the planner samples candidate action sequences, evaluates them with the learned score S, takes a top-ρ elite set, and changes the proposal parameters. This creates a feedback channel from model error to future sampling. A static proposal does not have this channel. Equal-budget static shooting uses the same total sample count but deliberately removes the feedback channel. That is why the static and equal-budget baselines are not auxiliary comparisons; they are the negative controls that define the paper.

The distinguishing empirical measurements are also optimizer-specific. The paper reports proposal drift, selected tail gap, elite pocket mass, elite model error, refit variance collapse, and sweep sensitivity across iteration counts. These quantities are not natural summary statistics for a one-shot selector. They are natural summary statistics for an adaptive sampling optimizer. If the paper is placed next to another manuscript about static candidate selection, Bayesian world-model ensembles, diffusion policies, memory retrieval, or language-symbolic planning, its evidence surface should still be recognizable: this paper is the one whose central variable is what CEM does between iteration t and iteration $t + 1$.

The paper also avoids claiming a universal "more search is bad" theorem. A generic candidate-selection paper might study monotonic degradation with candidate count or a selection bias theorem. This paper instead studies a conditional mechanism: if model-error pockets are elite-enriched, an elite-refit optimizer can increase proposal mass in those pockets. This is why the theoretical statement is a pocket-mass identity, not an order-statistic bound. The identity is simple, but it names the right failure mode for this repository: adaptation can amplify learned-model optimism.

The writing policy for v4 is therefore strict. Phrases such as "best of n" or "candidate selection" may appear only as descriptive background, not as the paper's contribution. The paper should foreground CEM, MPC, learned world models, elite refits, proposal drift, model-error pockets, and repair probes. If a reviewer skims only the abstract, introduction, figures, and appendix headings, they should see a coherent CEM-specific mechanism paper rather than a wrapper around a shared template.

APP.19 EVIDENCE TIERS AND CLAIM POLICY

The repository separates claims into evidence tiers. This is a deliberate submission policy rather than an afterthought. The paper's strongest results come from a controlled pocket world where the true utility, learned score, model-error pocket, and selected-tail gap can all be inspected. The paper's second tier is a sparse learned-ensemble bridge. The third tier is a sweep and short closed-loop sanity check. The unsupported tier is benchmark-scale transfer.

Tier 1: controlled mechanism evidence. Tier 1 supports statements about what happens in the controlled pocket world. It is valid to say that vanilla CEM has regret 11.64 in the verified run, that static proposal has regret 6.16, that equal-budget static proposal has regret 7.99, and that the best repair reduces regret by 11.43 points. It is also valid to say that selected tail gap and regret are strongly coupled in this benchmark, with correlation 0.99. The controlled setting is designed to make causal attribution legible; it is not designed to mimic the full visual-control stack of PlaNet or Dreamer.

Tier 2: sparse learned-ensemble bridge. Tier 2 supports cautious statements about learned-model optimism outside the analytic pocket construction. The sparse learned-ensemble experiment replaces the hand-specified hallucinated reward with a small bootstrapped learned model under sparse coverage. It supports the statement that calibrated learned CEM reduces regret from 4.64 to 2.96 in this bridge experiment. It does not support statements about high-dimensional latent dynamics, image observations, or broad model-based RL generalization.

Tier 3: stress and sanity evidence. Tier 3 includes the population/iteration/elite-fraction sweep and the short closed-loop test. The sweep supports statements about robustness within the controlled family: across 360 sweep rows, combined repair reduces mean regret from 9.08 to 2.33. The closed-loop test supports the narrower statement that the repaired planner improves a short receding-horizon return from -2.55 to 3.94. The closed-loop result is not a replacement for long-horizon MPC benchmarks.

Tier 4: unsupported external claims. Claims about PlaNet-scale, Dreamer-scale, MuJoCo-scale, or real-robot transfer are unsupported in v4. They may be valuable future work, but they are not allowed to migrate into the abstract, conclusion, or contribution bullets. The claim audit exists to preserve this boundary.

This tiering keeps the paper honest while still making it strong. Reviewers are more likely to trust a submission that states exactly where its evidence ends. The intended contribution is not that a small controlled benchmark proves all of MPC. The intended contribution is that a small controlled benchmark can expose an adaptive optimizer failure mode that final-return tables often hide.

APP.20 EXPERIMENT MATRIX AND ACCEPTANCE CRITERIA

Table 5 records the v4 experiment matrix as a reviewer-facing checklist. The point is to make clear which empirical role each artifact plays. A paper can be rejected for having many experiments that all answer the same question. This paper instead assigns each experiment a different inferential job.

The acceptance criteria are intentionally stronger than merely producing a plot. The controlled ranking must include static proposals because otherwise the paper cannot separate model-error exploitation from adaptive amplification. The trace progression must include iteration-wise diagnostics because otherwise the paper cannot show that the refit loop is involved. The repair ranking must disclose which repairs spend labels because otherwise the paper would blur deployable repairs with oracle-assisted probes. The sweep must vary iteration count because the mechanism specifically depends on repeated refits.

APP.21 FAILURE CASE NARRATIVES

The controlled benchmark can be understood through three failure narratives. The first is a benign static failure. A static proposal samples a fixed batch, the learned model overestimates a bad trajec-

Experiment	Primary question	Acceptance criterion
Controlled planner ranking	Does adaptive CEM underperform non-adaptive proposals in the pocket world?	Vanilla CEM must be worse than static and equal-budget static baselines, and the result must be visible in the v4 ranking figure.
Trace progression	Does the optimizer move toward the selected error tail across iterations?	Selected tail gap, elite pocket mass, and proposal drift should rise together for vanilla CEM traces.
Repair ranking	Do interventions aimed at score optimism or refit concentration reduce regret?	At least one score-side repair and one refit-side repair must be evaluated, and pilot-label use must be disclosed.
Population/iteration sweep	Is the repair pattern tied to a single hyperparameter setting?	Combined repair should improve mean regret across the sweep table, not only in the main setting.
Sparse learned ensemble	Does a learned sparse model show the same qualitative problem?	Calibrated learned CEM should improve over vanilla learned CEM, with the claim labeled as a bridge.
Closed-loop sanity check	Does replanning immediately erase the open-loop issue?	The repaired planner should improve short receding-horizon return, while the paper admits the horizon remains small.

Table 5: V4 experiment matrix. Each row has a different inferential role.

tory, and the planner selects it. This is a normal model-exploitation problem. It is important, but it is not the new mechanism here.

The second narrative is adaptive concentration. In this case, the first CEM iteration discovers a few overoptimistic candidates in the pocket. Because those candidates are top-scoring under the learned model, they enter the elite set. The refit moves the proposal mean or variance toward their action statistics. The next iteration samples more densely around them, which raises the chance of drawing even more pocket candidates. The optimizer has started to train its own proposal distribution on model error.

The third narrative is variance lock-in. Once the proposal variance collapses around the pocket, later iterations have fewer opportunities to escape. This does not require every sample to be in the pocket. It only requires the proposal to spend enough mass near the selected error tail that the best learned-model score in later iterations is again a pocket trajectory. This is why conservative temperature can help: it keeps the proposal from collapsing too quickly. It is also why temperature alone is not a guarantee. If the score model remains strongly optimistic, high variance may simply keep rediscovering the same tail.

These narratives explain why final return is insufficient. A final-return table can show that vanilla CEM is worse than a repair, but it does not show whether the optimizer adapted toward the wrong signal. The v4 trace figure does show that. It makes the failure dynamic: selected tail gap rises, elite pocket mass rises, and the proposal moves.

App.22 Alternative Hypotheses

The paper should be judged not only by whether its favored explanation fits, but by whether plausible alternatives are addressed.

Hypothesis 1: vanilla CEM is worse only because it uses more samples. The equal-budget static baseline directly targets this explanation. It spends the same total sample budget without refitting between iterations. If sample count alone caused the failure, equal-budget static shooting should be as bad as vanilla CEM. In the controlled run, vanilla CEM remains worse. This does not prove sample count never matters. It shows that repeated adaptation is an additional factor in this benchmark.

Hypothesis 2: vanilla CEM is worse only because its final sample set is unlucky. The trace diagnostics make this unlikely. The failure is not visible only at the final selected action. Iteration-wise selected tail gap and elite pocket mass increase during the refit process. This pattern is more consistent with a systematic refit effect than with a single unlucky final sample.

Hypothesis 3: all planners fail equally under model error. The static, equal-budget static, and repair baselines reject this explanation inside the controlled benchmark. They all use the same broad learned-model setting, but their true-regret outcomes differ substantially. The difference is therefore not reducible to "the model is wrong." The optimizer and repair interface matter.

Hypothesis 4: repairs work only because they secretly use true labels. The repository includes label-leakage tests, and the paper explicitly marks pilot-label calibration as the repair that spends real labels. Uncertainty pessimism, disagreement veto, diversity floor, conservative temperature, and shadow-realism scoring do not call true evaluation labels during planning. This distinction matters because label-spending repairs and label-free probes answer different deployment questions.

Hypothesis 5: the mechanism is an artifact of hand-designed optimism. The sparse learned-ensemble bridge weakens this objection but does not remove it. The bridge shows that learned sparse-region bias can produce a similar qualitative pattern. However, the paper still admits that high-dimensional transfer is future work. This is the correct boundary: the analytic pocket is an instrument for causal clarity, and the learned ensemble is a bridge, not a final external-validation suite.

Hypothesis 6: the result disappears under receding-horizon planning. The short closed-loop evaluation tests whether immediate replanning erases the problem. It does not. Combined repair improves short-horizon return over vanilla CEM. The result is intentionally labeled as a sanity check because longer horizons and richer starts remain untested.

APP.23 REPAIR COMPONENT INTERPRETATION

The repair suite is not a proposal for a single new algorithm. It is a set of interventions placed at different locations in the failure chain.

Score-side repairs. Uncertainty pessimism and disagreement veto operate before elite selection. They ask whether optimistic model-score artifacts can be downweighted or filtered before they become the training signal for the next proposal. Their success supports the claim that score optimism is part of the failure. Their limitations are equally important: if all ensemble members share the same bias, disagreement can be low, and a disagreement veto may not catch the pocket.

Refit-side repairs. Conservative temperature and elite diversity floors operate on the adaptation step. They ask whether slowing concentration or preventing elite collapse can reduce the damage even when the learned score remains imperfect. These repairs are closest to the theorem's mechanism, because the theorem is about how conditioning on elites changes the proposal. Their mixed performance is useful rather than embarrassing: it shows that simply adding diversity is not enough when all diverse candidates share the same optimism source.

Calibration repairs. Pilot-label calibration uses a small set of true labels to fit an optimism penalty. It performs strongly, but it is not label-free. The correct reading is that targeted real feedback can interrupt the pocket; the incorrect reading is that the paper has solved deployment without supervision. The manuscript keeps this distinction visible because reviewers will notice if the strongest repair quietly spends information unavailable to the other methods.

Combined repair. The combined repair should be interpreted as a stress intervention: it checks whether multiple weak signals aimed at the mechanism can work together across the sweep and closed-loop test. It should not be presented as the final method. If a future version wants to propose a deployable algorithm, it should isolate which components are necessary, tune them on a validation protocol, and test them on external benchmarks.

APP.24 STATISTICAL REPORTING POLICY

The experiments in v4 are small enough that overclaiming statistical precision would be misleading. The paper reports means and effect sizes because the goal is mechanism exposure, not leaderboard comparison. When the paper says that vanilla CEM has higher regret than static proposal, the support is the full controlled artifact and the v4 planner ranking, not a claim of universal dominance across all possible pocket worlds.

For future external benchmarks, the reporting policy should become stricter. The minimum benchmark-scale report should include at least three random seeds per environment, confidence intervals or paired seed differences, planner budget matching, model-training seed disclosure, and a separate validation set for repair hyperparameters. It should also report negative results. If a repair helps in the controlled pocket world but fails in a visual-control benchmark, that failure would be scientifically valuable because it would separate mechanism diagnosis from deployable performance.

The v4 paper's present statistics are best read as deterministic artifacts of an inspectable experimental system. The cached evidence script records counts, tables, and summary values so that the numbers in the manuscript can be checked mechanically. This is more important for v4 than adding decorative significance tests. The central question is whether the mechanism is visible, whether the baselines isolate adaptation, and whether the claims stay within the measured scope.

APP.25 HYPERPARAMETER SENSITIVITY PROTOCOL

The sweep in v4 varies population, iteration count, elite fraction, and seed. Those are the right first axes because the mechanism depends on adaptive sampling. Population controls how often rare optimistic candidates are found. Iteration count controls how many times elite selection can affect future sampling. Elite fraction controls the aggressiveness of the refit. Seed controls the random candidate stream and learned-model sampling variation.

Other axes remain future work. Horizon can change both the probability of entering a pocket and the severity of compounding model error. Action dimension can make diagonal Gaussian refits less expressive and can alter how quickly variance collapses. Initial variance can determine whether the pocket is discoverable at all. Momentum in the CEM update can either smooth noisy elite sets or preserve drift toward the pocket. Repair coefficients can trade off pessimism against useful exploration.

The intended protocol for future sensitivity is sequential and RAM-light. First, run a coarse grid over one axis while holding the others fixed. Second, select suspicious regimes where vanilla CEM and static proposals diverge. Third, run paired repair comparisons only in those regimes. Fourth, write compact CSV summaries after each run rather than keeping all trajectories in memory. This protocol respects the user's compute constraint while still expanding experimental scope.

The current v4 sweep is enough to show that the main result is not a single population or iteration setting. It is not enough to claim full hyperparameter invariance. That distinction should remain visible in the paper because it is exactly the kind of nuance a harsh reviewer will ask for.

APP.26 IMPLEMENTATION AUDIT

The implementation has four correctness boundaries. First, planner scoring and true evaluation must remain separate. A planner may use the learned score, uncertainty, disagreement, support proxies, or explicitly supplied pilot labels. It may not call the true utility for arbitrary candidates during planning. Second, static and equal-budget static baselines must use comparable candidate budgets. Otherwise the paper could mistake budget differences for adaptation effects.

Third, trace metrics must be computed from the planner's actual sampled candidates and elite sets. A trace generated after the fact from a different proposal would not test the refit loop. Fourth, v4 paper numbers must be derived from repository artifacts, not typed manually. The macros file written by `experiments/v4_protocol_evidence.py` is a practical guardrail: it imports effect sizes into LaTeX from the same summary JSON used by the claim audit.

The claim audit complements the implementation tests. Unit tests can check that functions behave locally. The claim audit checks that paper-level claims cite evidence and do not silently expand beyond the experiments. Both are needed for submission readiness. A repository can pass unit tests while still making unsupported claims in prose; conversely, cautious prose cannot rescue a planner implementation that leaks labels.

APP.27 RAM-LIGHT EXECUTION STRATEGY

The experiments are designed to be run without high memory pressure. The v4 pipeline uses committed full-run artifacts and writes compact derived ledgers. The cached evidence script streams through CSV-style tables, aggregates with small data frames, writes summary JSON, and emits figures. It does not keep a large replay buffer, image dataset, or model checkpoint ensemble in memory.

For future expansion, the same policy should hold. Heavy benchmark runs should be split by environment, planner, seed, and hyperparameter cell. Each job should write an intermediate result file immediately. Aggregation should read only the columns needed for each table or figure. If a learned ensemble is large, member predictions should be summarized into means, variances, and disagreement metrics before paper-level aggregation. Sequential execution is preferred over parallel execution when RAM is the bottleneck.

The important point is that RAM-light does not mean weak experiments. It means the experiment design treats storage as the synchronization boundary. The user wants maximum rigor without letting the machine become unstable. This paper's v4 audit follows that pattern by reusing full CPU artifacts, deriving mechanism-specific summaries, and building the manuscript from cached values.

APP.28 RESULT READING GUIDE

A reviewer should read the v4 results in a specific order. First, inspect the controlled planner ranking. It establishes that vanilla CEM is worse than static proposals in the benchmark where the true objective is known. Second, inspect the trace progression. It establishes that the failure develops across elite refits rather than appearing only as a final selection accident. Third, inspect the selected-tail-gap plot. It establishes that the selected optimism diagnostic is coupled to regret inside the controlled system.

Only after those three checks should the repair tables be interpreted. If the mechanism were not visible, repair improvements would be hard to attribute. Once the mechanism is visible, the repair table asks which interventions interrupt it. The best repair is not automatically the paper's main contribution. The main contribution is the combination of mechanism, negative controls, diagnostics, and bounded repair evidence.

The learned-ensemble and closed-loop panels should be read last. They are bridges outward from the controlled setting. They prevent the paper from being only a toy analytic story, but they do not carry the weight of external benchmark validation. This ordering protects the manuscript from the most dangerous overclaim: using small bridge experiments to imply broad transfer.

APP.29 AREA-CHAIR DECISION SIMULATION

An area chair looking for reasons to reject may start with the synthetic benchmark. The strongest response is not to pretend that the benchmark is realistic. The strongest response is that the benchmark is an instrument. It has a known error pocket, known true return, and measurable proposal drift, which makes a hidden optimizer-model feedback loop visible. The paper then adds a learned-ensemble bridge and closed-loop sanity check without pretending they are benchmark-scale proof.

A second rejection path is novelty. CEM, uncertainty penalties, and model-based RL repair ideas are all well known. The paper's novelty is the diagnostic framing of CEM refits as an adaptive model-error amplification channel, together with negative controls that separate sample-count exposure from repeated refit. The repair methods are intentionally not sold as new algorithms. This should be stated plainly because overselling repairs would make the paper easier to reject.

A third rejection path is duplicate-template suspicion. The v4 paper should survive that attack because its title, theorem, evidence, and appendix revolve around CEM elite refits. The paper does not share a generic selection theorem as its main contribution. It has optimizer-specific diagnostics and baselines: equal-budget static shooting, proposal drift, elite pocket mass, variance collapse, and iteration sweeps.

A fourth rejection path is insufficient evidence. The v4 response is to be transparent. The paper is no longer a short draft: it includes a full evidence ledger, figures, sweep, bridge experiment, closed-loop sanity check, claim trace, and implementation audit. It still does not claim benchmark-scale transfer. This makes the paper submission-ready as a bounded mechanism paper, not as a final general MPC benchmark study.

APP.30 SUBMISSION-READINESS CHECKLIST

Before v4 is treated as final, the following checks must pass:

1. The final PDF has at least 25 pages and is written to `paper/final/` with a v4 filename.
2. The visible Desktop copy has the same SHA-256 hash as the repository final PDF.
3. The source map row names the v4 Desktop PDF, the local source folder, and the GitHub repository.
4. The old root draft PDF is not the submission artifact.
5. The unversioned compiled `paper/iclr_submission/main.pdf` is not the artifact a fresh agent should use as final.
6. `experiments/v4_protocol_evidence.py` regenerates the v4 summary, CSV ledgers, figures, and macros.
7. The v4 summary counts match the manuscript macros.
8. `scripts/run_claim_audit.py` passes against `docs/claims.json`.
9. The v4 audit script checks page count, hashes, evidence values, stale artifact references, and source-map consistency.
10. Unit tests pass.
11. Python files compile.
12. The LaTeX log contains no undefined references, fatal errors, or citation failures.
13. A visual check samples pages 1, 5, 10, 15, 20, and 25.
14. The Git working tree is clean after commit.
15. The pushed GitHub main branch matches the local commit.

This checklist is intentionally concrete. It turns ”submission ready” into a set of observable repository states rather than a mood.

APP.31 HOW TO REJECT THIS PAPER CORRECTLY

A correct rejection would not say that the paper proves nothing because the benchmark is controlled. The paper already agrees that the benchmark is controlled. A correct rejection would say that the mechanism is interesting but the venue requires external benchmark validation, or that the learned-ensemble bridge is too small for the claims the venue expects.

A correct rejection would not say that uncertainty penalties are not new. The paper already says they are mechanism probes, not new algorithms. A correct rejection would say that the diagnostic contribution is not enough without a new repair method or broader empirical scope.

A correct rejection would not say that CEM is always bad. The paper does not claim that. A correct rejection would say that the paper needs to better characterize when the elite-enrichment condition holds in realistic learned models.

Planner family	Adaptive refit?	What the comparison isolates
Random shooting	No	Baseline tail exposure from an uninformed fixed proposal.
Static proposal	No	Learned-score exploitation without repeated proposal updates.
Equal-budget static	No	Same total candidate count as multi-iteration CEM, but no elite-to-proposal feedback.
Vanilla CEM	Yes	Candidate exposure plus elite-conditioned proposal updates.
Score-side repaired CEM	Yes	Whether uncertainty, disagreement, or calibration can prevent bad candidates entering the elite training signal.
Refit-side repaired CEM	Yes	Whether slowing or diversifying the update can reduce concentration after elite selection.

Table 6: Budget-matching logic. The equal-budget static baseline is the key negative control for separating sample count from adaptive refit.

These distinctions matter because they sharpen the paper. By naming the valid rejection paths, the manuscript avoids wasting reviewer attention on claims it does not make. It also helps the authors decide what v4 would need: external continuous-control benchmarks, larger learned models, longer closed-loop horizons, and repair-component validation.

APP.32 HOW TO ACCEPT THIS PAPER CORRECTLY

A correct acceptance would view the paper as a mechanism and audit contribution. The value is not that the controlled pocket world is itself a benchmark of practical robotics. The value is that it isolates a feedback loop that can be hidden inside learned-model MPC: model-scored elites become the training signal for the next proposal, so sparse optimism can become self-reinforcing.

A correct acceptance would also value the negative controls. Equal-budget static shooting is a clean comparison because it preserves sample count while removing refit adaptation. The trace metrics then show that the optimizer changes its distribution toward the selected error tail. Together, these pieces make a stronger mechanism case than a final-return table alone.

Finally, a correct acceptance would value the claim boundary. The paper does not convert controlled evidence into broad benchmark claims. It includes a learned-ensemble bridge, sweep, closed-loop sanity check, and explicit unsupported-claim registry. That restraint is part of the contribution because it makes the audit reusable: future work can add benchmark rows without rewriting the mechanism.

APP.33 PLANNER BUDGET MATCHING DETAILS

Budget matching is central to the paper because the mechanism is adaptive refit, not total search effort. The controlled benchmark therefore compares vanilla CEM against two static baselines. The first static baseline samples from a fixed proposal once. The equal-budget static baseline samples the same total number of candidates that multi-iteration CEM would sample, but it does not refit the proposal between batches. This preserves tail exposure while removing adaptive concentration.

The distinction matters because a learned model with a high-error tail can hurt any optimizer that samples enough candidates. A large static proposal may find a bad overoptimistic candidate. That is not disputed. The paper asks a sharper question: after such candidates appear in an elite set, does refitting the proposal amplify their future probability? Equal-budget static shooting is the minimal negative control for this question.

The paper should not compare vanilla CEM only against random shooting. That would be too weak because random shooting differs in both proposal structure and adaptation. It should not compare only against a low-budget static proposal. That would leave open the possibility that CEM fails merely because it sees more candidates. The current v4 comparisons are deliberately arranged so that each baseline removes one possible confound.

The sweep extends the same logic. Population changes exposure to rare tails. Iteration count changes the number of times elite selection can feed back into the proposal. Elite fraction changes how concentrated the feedback signal is. The repair-gain table is useful because it asks whether the combined repair helps in more than one budget regime. The result should still be read within the controlled family, but it is stronger than a single fixed-budget run.

APP.34 TRACE METRIC DEFINITIONS

Trace metrics are the paper's main defense against the criticism that the result is just a final-score anomaly. The following definitions specify what each trace statistic means.

Selected tail gap. For the selected candidate a_{sel}, the selected tail gap is $S(a_{\mathrm{sel}}) - U(a_{\mathrm{sel}})$. It measures how much the learned model overestimates the action sequence the planner actually selects. This is not a deployment metric in settings where U is unavailable at planning time. It is an audit metric used after evaluation or in controlled benchmarks.

Elite model error. Elite model error averages $S(a) - U(a)$ over candidates selected into the elite set. This statistic asks whether the update signal itself is contaminated by model optimism. A high final selected-tail gap is concerning; high elite model error is more mechanism-specific because elites are what define the next proposal.

Elite pocket mass. Elite pocket mass is the fraction of elite candidates entering the known model-error pocket. In the controlled world, the pocket is explicitly defined, so this metric is directly observable. In external benchmarks, analogous metrics would require delayed true labels, support diagnostics, or human defined risky regions.

Proposal drift. Proposal drift measures how far the current proposal parameters have moved from the initial proposal. In diagonal Gaussian CEM, this can include mean movement and variance change. Drift alone is not bad: a good optimizer should move toward good actions. Drift becomes evidence for the mechanism when it co-occurs with elite pocket enrichment and rising selected tail gap.

Variance collapse. Variance collapse measures whether the proposal loses coverage as it adapts. Collapse is useful when the optimizer has found a truly good basin. It is harmful when the basin is a learned-model artifact. Conservative temperature targets this metric by preserving exploration longer, but high variance can also reduce performance if it prevents exploitation.

The metrics should be interpreted jointly. Any one statistic can be misleading. A high selected tail gap without proposal drift could be a one-shot selection failure. Proposal drift without elite model error could be normal convergence. Elite pocket mass without regret could mean the pocket is not actually harmful under the true utility. The paper's mechanism claim is strongest when these measures move together, as they do in the v4 vanilla CEM trace.

APP.35 BOUNDARY CONDITIONS

The mechanism requires elite enrichment of model-error pockets. If a learned model's errors are small, symmetric, or uncorrelated with the score ranking, CEM should not amplify them in the same way. If the true optimum lies inside the same region that the learned model favors, proposal drift is useful rather than harmful. If the proposal family cannot represent the pocket geometry, refits may not concentrate pocket mass even when elites are enriched. If the planner uses strong support constraints or accurate uncertainty estimates, bad pocket candidates may never enter the elite set.

These boundary conditions are not weaknesses of the theorem. They are the theorem's content. The identity says that pocket mass increases when the pocket is elite-enriched. It does not say every

Logged quantity	Frequency	Purpose
Candidate learned score	Every CEM iteration	Identifies the ranking signal used for elite selection.
Candidate uncertainty or disagreement	Every CEM iteration	Tests whether overoptimistic candidates are also uncertain.
Elite indices and scores	Every CEM iteration	Reconstructs the update signal.
Proposal mean and variance	Every CEM iteration	Measures drift and collapse.
Executed action and realized return	Every environment step	Connects planner choice to true outcome.
Delayed true return for audited candidates	Audit subset	Estimates selected tail gap when full labeling is impossible.
Support or density proxy	Every candidate or audit subset	Tests whether elites move out of the data-support region.
Repair coefficients and random seeds	Every run	Enables exact reproducibility and paired comparisons.

Table 7: Minimum logging schema for external benchmark transfer. The schema is designed to preserve the mechanism audit, not only final performance.

learned model creates elite-enriched pockets. This is why the paper should avoid universal language. The correct claim is conditional and diagnostic: when elite enrichment occurs, the refit loop can turn model error into a self-reinforcing sampling bias.

The mechanism can also fail to appear if CEM hyperparameters are too mild. With only one iteration, CEM is close to one-shot selection. With a very large elite fraction, refits may be too diffuse to concentrate strongly. With very high minimum variance, the proposal may not lock in. With very low population, the optimizer may rarely sample the pocket at all. These are not contradictions. They describe the operating region in which adaptive refit is most likely to matter.

Conversely, the mechanism may become more severe in high-dimensional systems when learned models have narrow but high-scoring artifacts, when online planning repeatedly revisits similar states, or when action-sequence proposals collapse around visually plausible but dynamically invalid rollouts. The v4 paper does not prove those cases, but it gives a vocabulary for testing them.

APP.36 EXTERNAL BENCHMARK LOGGING SCHEMA

Future benchmark work should not only report episode return. It should log the optimizer-model interaction at the candidate level. Without that logging, a researcher may observe that a repair helps but be unable to determine whether it helped by reducing model-error exploitation, preserving proposal diversity, or simply changing the effective search budget.

This schema is intentionally heavier than ordinary benchmark logging but still RAM-light. Candidate-level logs can be written per seed and compressed after aggregation. The paper-level analysis does not need to hold all candidates in memory at once. It only needs enough information to reconstruct elite selection, proposal drift, uncertainty, support, and realized outcome for a sample of audited candidates.

External benchmarks should also separate planning budget from model-training budget. A repair that improves return by slowing the planner may change wall clock cost. A repair that requires additional labels changes data cost. A repair that requires an ensemble changes memory and training cost. The v4 paper does not hide this issue; it treats pilot labels and ensemble disagreement as different resources.

APP.37 REPRODUCIBILITY RECIPE

A fresh agent or reviewer should be able to reproduce the v4 artifact without guessing which file is authoritative. The source folder is the repository root. The paper source is `paper/iclr_submission/main.tex`. The v4 build script is `scripts/build_v4_paper.py`. The paper-facing evidence cache is `results/v4_protocol_evidence/`. The final repository PDF is stored under `paper/final/`. The visible Desktop PDF has the same v4 filename.

The reproducibility procedure has five steps. First, run the existing tests to check planner and repair invariants. Second, run the claim audit against `docs/claims.json`. Third, run the v4 protocol evidence script and the Gymnasium card script to regenerate summary values and figures. Fourth, rebuild the paper so the macros and figures are synchronized. Fifth, run the v4 audit to check page count, hashes, source-map consistency, Gymnasium gates, and stale artifacts.

This recipe deliberately avoids requiring expensive reruns for every fresh session. The full CPU artifacts are already committed. The v4 script is a paper-facing aggregation pass over those artifacts. A reviewer who wants to rerun the full experiments can use the full-run scripts, but a reviewer who only wants to check paper consistency can regenerate the v4 layer quickly.

APP.38 REVIEWER STRESS QUESTIONS

What if CEM is simply badly tuned? The sweep partially addresses this by varying population, iteration count, and elite fraction. More importantly, the paper is not claiming that every CEM configuration fails. It claims that adaptive refits can amplify model-error elites under identifiable conditions. If a different tuning avoids elite enrichment, that is consistent with the mechanism.

What if the static baseline is too strong or too weak? The paper includes both ordinary static proposal and equal-budget static proposal. If the ordinary static baseline were too low-budget, equal-budget static would correct that confound. If static proposal were too strong because it avoids adaptation, that is exactly the point: removing the feedback channel can be protective in this controlled pocket world.

What if the true optimum is misestimated? The controlled benchmark uses a strengthened true-oracle estimate for regret. The important comparisons are paired within the same evaluation protocol. A future version could add denser oracle sweeps, but the v4 conclusion does not depend on comparing against a hidden benchmark leaderboard. It depends on relative regret and trace behavior under the same controlled utility.

What if the learned-ensemble bridge is too small? It is small, and the paper says so. The bridge is included to check whether the diagnostic vocabulary survives when optimism comes from learned sparse coverage. It is not used to claim broad transfer. A reviewer may still demand larger benchmarks, but that is a scope decision rather than a contradiction.

What if repairs reduce regret by being overly conservative? That is possible for some repairs. The paper's repair interpretation therefore focuses on mechanism probes, not deployment guarantees. A conservative repair that reduces regret in the pocket world is evidence that preventing overoptimistic elite concentration matters; it is not evidence that the same repair will optimize return in all domains.

What if selected tail gap is unobservable in real systems? Selected tail gap is an audit metric. In real systems it may require delayed outcomes, simulator labels, offline evaluation, or an audited candidate subset. The metric is still valuable because it names what must be approximated when a planner is suspected of exploiting learned-model artifacts.

What if the paper is too cautious? The caution is intentional. A mechanism paper can be strong without claiming more than it measured. Overstating transfer would make the submission easier to attack. The v4 version instead aims to be hard to reject on internal validity: the mechanism is defined, baselines isolate it, diagnostics measure it, repairs probe it, and unsupported claims are kept out.

APP.39 FUTURE BENCHMARK ROADMAP

The v4 manuscript is submission-ready as a bounded mechanism paper with exact Gymnasium transition-card stress tests. The most valuable next expansion would be a continuous-control benchmark with learned dynamics and cheap true evaluation. This would allow candidate-level audit labels for a subset of planned trajectories. The second target should be a latent-dynamics benchmark where the planner operates in learned latent space and true returns are observed after execution. The third target should compare CEM against gradient-based and hybrid MPC optimizers.

For each target, the same questions should be asked. Does vanilla CEM select high learned-score candidates with poor true outcomes? Are those candidates uncertain or out of support? Does elite selection enrich them? Does proposal drift move toward them? Does equal-budget static shooting fail less or differently? Which repair interrupts the chain, and what resource does that repair spend?

The roadmap should not be used to weaken v4. A paper can be complete while still having future work. This version's job is to establish the mechanism cleanly, add low-resource standard-environment stress cards, and make the audit contract reproducible. Future benchmark work should test how often the mechanism matters in larger continuous-control systems.

APP.40 ARCHIVE CONTRACT

The final v4 artifact is `best-of-n-MPC-CEM-latent-world-models-v4.pdf`. The build copies this PDF to the visible Desktop and to `paper/final/`. Older root drafts and unversioned compiled PDFs are not the submission artifact. This matters because a fresh session should be able to identify the correct source folder, GitHub repo, and final PDF without reading the whole history.

The archive contract also protects the claim boundary. The final repo must contain the v4 evidence summary, CSV ledgers, figures, macros, claim audit, and final PDF. If a later version adds benchmark-scale evidence, it should create a new versioned evidence layer and final PDF rather than overwriting v4 claims in place.

APP.41 LARGE-LANGUAGE-MODEL USAGE DISCLOSURE

Codex, an OpenAI language model accessed through a local coding-agent interface, assisted with repository inspection, evidence-script generation, LaTeX editing, build verification, and claim-boundary checks. Numeric claims come from repository artifacts and generated ledgers, not from model invention. The human authors remain responsible for scientific claims, artifact selection, and final submission decisions.